US States Health Analysis

October 22, 2026

1 CATEGORICAL HEALTH ANALYSIS OF US STATES

[298]:
```
import re
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import plotly.express as px
import seaborn as sns
%matplotlib inline
from functools import reduce
```

1.1 Creating dataframes for all the csv files to be Analysed and merged after cleaning

[299]:
```
pd.set_option("display.max.columns", None)
pd.set_option('display.precision', 2)
```

[300]:
```
diverse_states_df = pd.read_csv('diverse_states.csv')
expensive_states_df = pd.read_csv('expensive_states_healthcare.csv')
health_care_costs_by_state_df = pd.
 ↪read_csv('healthcare_costs_share_of_income_by_state.csv')
healthcare_spending_per_person_df = pd.
 ↪read_csv('healthcare_spending_per_person_by_state.csv')
healthiest_states_df = pd.read_csv('healthiest_states.csv')
life_expectancy_by_state_df = pd.read_csv('life_expectancy_by_state.csv')
obesity_rate_df = pd.read_csv('obesity_rates_in_US_States.csv')
```

[301]:
```
obesity_rate_df.rename(columns={'State or Territory': 'State'}, inplace=True) #␣
 ↪Changed column name for merging
```

1.2 Creating a sorted list of dataframes to prepare for merging

[302]:
```
dfs = [diverse_states_df, expensive_states_df, health_care_costs_by_state_df,␣
 ↪healthcare_spending_per_person_df,
       healthiest_states_df, life_expectancy_by_state_df, obesity_rate_df]
```

1.2.1 Dropping duplicate column on the dataframes to prevent suffix errors while merging

[303]:
```
dfs = [df.drop('Rank', axis=1) for df in dfs]
```

1.2.2 Merging on the State column

[304]:
```
merged_df = reduce(lambda  left, right: pd.merge(left, right, on=['State'],␣
↪how='outer', suffixes=('_left', '_right')), dfs)
```

1.2.3 Merging States Abbreviations

[305]:
```
abbr_df = pd.read_csv('US States abbreviations.csv')
```

[306]:
```
abbr_df.head()
```

[306]:
```
         State Abbreviation
0      Alabama           AL
1       Alaska           AK
2      Arizona           AZ
3     Arkansas           AR
4   California           CA
```

[307]:
```
abbr_df.tail()
```

[307]:
```
            State Abbreviation
45       Virginia           VA
46     Washington           WA
47  West Virginia           WV
48      Wisconsin           WI
49        Wyoming           WY
```

[308]:
```
df_merged = pd.merge(merged_df, abbr_df, on='State', how='outer')
```

[309]:
```
df_merged = df_merged.reindex(['Abbreviation', 'State', 'Diversity Index (%)',␣
↪'Healthcare Cost Index (U.S. = 100)', 'Health Care as % of Household Income',
                               'Per-Capita Health Care Spending', 'Overall␣
↪Score 2025', 'Adult Obesity Rate (2024)', 'Life Expectancy (Years)'], axis=1)
```

[ ]:
```

```

1.2.4 Viewing merged dataframe

[310]:
```
df_merged.head()
```

[310]:
```
  Abbreviation       State  Diversity Index (%)  \
0           AL     Alabama                 53.1
1           AK      Alaska                 62.8
```

```
2           AZ     Arizona                61.5
3           AR    Arkansas                49.8
4           CA  California                69.7

   Healthcare Cost Index (U.S. = 100) Health Care as % of Household Income  \
0                                85.6                                 7.32%
1                               130.7                                10.08%
2                                98.8                                 6.81%
3                                85.8                                 7.67%
4                               111.1                                 5.64%

  Per-Capita Health Care Spending  Overall Score 2025  \
0                         $7,980               -0.75
1                        $14,044               -0.13
2                         $8,556               -0.06
3                         $8,562               -0.83
4                        $11,054                0.10

  Adult Obesity Rate (2024)  Life Expectancy (Years)
0                     38.9%                     73.8
1                     34.0%                     75.8
2                     33.3%                     76.7
3                     38.9%                     73.9
4                     29.1%                     79.3
```

[311]:
```
df_merged.tail()
```

[311]:
```
   Abbreviation                          State  Diversity Index (%)  \
52          NaN               Washington, D.C.                 67.2
53           WV                  West Virginia                 20.2
54           WI                      Wisconsin                 37.0
55           WY                        Wyoming                 32.4
56          NaN  U.S. State and Territory Average                  NaN

    Healthcare Cost Index (U.S. = 100) Health Care as % of Household Income  \
52                                 NaN                                  NaN
53                                89.5                                9.14%
54                               105.3                                6.76%
55                                97.0                                7.84%
56                                 NaN                                  NaN

   Per-Capita Health Care Spending  Overall Score 2025  \
52                             NaN                 NaN
53                        $12,055               -0.73
54                        $10,079                0.16
55                         $9,640               -0.04
56                             NaN                 NaN
```

```
    Adult Obesity Rate (2024)  Life Expectancy (Years)
52                        NaN                      NaN
53                      41.4%                     72.2
54                      37.4%                     78.1
55                      32.5%                     76.8
56                      34.1%                      NaN
```

1.2.5 Renaming some columns for Clarity

[312]:
```
df_merged.rename(columns={'Overall Score 2025': 'Health Score', 'Adult Obesity␣
↪Rate (2024)': 'Adult Obesity (%)',
                          'Per-Capita Health Care Spending':'Per-Capita Health␣
↪Care Spending ($)'},
                 inplace=True) # Changed column names to properly reflect␣
↪column data
```

[313]:
```
df_merged.head()
```

[313]:
```
  Abbreviation       State  Diversity Index (%)  \
0           AL     Alabama                 53.1
1           AK      Alaska                 62.8
2           AZ     Arizona                 61.5
3           AR    Arkansas                 49.8
4           CA  California                 69.7

   Healthcare Cost Index (U.S. = 100) Health Care as % of Household Income  \
0                                85.6                                 7.32%
1                               130.7                                10.08%
2                                98.8                                 6.81%
3                                85.8                                 7.67%
4                               111.1                                 5.64%

  Per-Capita Health Care Spending ($)  Health Score Adult Obesity (%)  \
0                             $7,980         -0.75             38.9%
1                            $14,044         -0.13             34.0%
2                             $8,556         -0.06             33.3%
3                             $8,562         -0.83             38.9%
4                            $11,054          0.10             29.1%

   Life Expectancy (Years)
0                     73.8
1                     75.8
2                     76.7
3                     73.9
4                     79.3
```

1.2.6 Removing %, $ and commas using regex from the column data for Visualization

[314]:
```
df_merged.replace(r"(^\$)|(%$)|(\,)", '', regex=True, inplace=True)
```

[314]:

	Abbreviation	State	Diversity Index (%)	\
0	AL	Alabama	53.1	
1	AK	Alaska	62.8	
2	AZ	Arizona	61.5	
3	AR	Arkansas	49.8	
4	CA	California	69.7	
5	CO	Colorado	52.3	
6	CT	Connecticut	55.7	
7	DE	Delaware	59.6	
8	NaN	District of Columbia	NaN	
9	FL	Florida	64.1	
10	GA	Georgia	64.1	
11	NaN	Guam	NaN	
12	HI	Hawaii	76.0	
13	ID	Idaho	35.9	
14	IL	Illinois	60.3	
15	IN	Indiana	41.3	
16	IA	Iowa	30.8	
17	KS	Kansas	45.4	
18	KY	Kentucky	32.8	
19	LA	Louisiana	58.6	
20	ME	Maine	18.5	
21	MD	Maryland	67.3	
22	MA	Massachusetts	51.6	
23	MI	Michigan	45.2	
24	MN	Minnesota	40.5	
25	MS	Mississippi	55.9	
26	MO	Missouri	40.8	
27	MT	Montana	30.1	
28	NE	Nebraska	40.8	
29	NV	Nevada	68.8	
30	NH	New Hampshire	23.6	
31	NJ	New Jersey	65.8	
32	NM	New Mexico	63.0	
33	NY	New York	65.8	
34	NC	North Carolina	57.9	
35	ND	North Dakota	32.6	
36	OH	Ohio	40.4	
37	OK	Oklahoma	59.5	
38	OR	Oregon	46.1	
39	PA	Pennsylvania	44.0	
40	NaN	Puerto Rico	NaN	
41	RI	Rhode Island	49.4	

42	SC	South Carolina	54.6
43	SD	South Dakota	35.6
44	TN	Tennessee	46.6
45	NaN	Tennessee*	NaN
46	TX	Texas	67.0
47	UT	Utah	40.7
48	VT	Vermont	20.2
49	NaN	Virgin Islands	NaN
50	VA	Virginia	60.5
51	WA	Washington	55.9
52	NaN	Washington D.C.	67.2
53	WV	West Virginia	20.2
54	WI	Wisconsin	37.0
55	WY	Wyoming	32.4
56	NaN	U.S. State and Territory Average	NaN

	Healthcare Cost Index (U.S. = 100)	Health Care as % of Household Income	\
0	85.6	7.32	
1	130.7	10.08	
2	98.8	6.81	
3	85.8	7.67	
4	111.1	5.64	
5	103.9	6.05	
6	108.9	6.89	
7	101.6	6.66	
8	118.7	NaN	
9	101.4	6.05	
10	99.6	6.62	
11	NaN	NaN	
12	113.1	8.00	
13	101.6	7.82	
14	103.0	7.71	
15	91.5	6.64	
16	94.3	6.74	
17	90.6	7.19	
18	92.4	7.46	
19	91.6	8.13	
20	114.7	9.30	
21	106.9	6.76	
22	131.8	8.00	
23	88.2	6.65	
24	111.5	6.09	
25	85.1	9.18	
26	98.0	7.43	
27	106.3	8.62	
28	105.6	6.87	
29	105.4	5.87	

30	106.5	6.42
31	115.8	5.81
32	101.9	9.07
33	108.9	7.98
34	108.4	8.78
35	107.7	8.08
36	96.5	6.68
37	98.1	7.51
38	115.3	9.32
39	95.1	6.80
40	NaN	NaN
41	104.1	6.09
42	97.6	6.13
43	104.1	8.60
44	88.7	6.77
45	NaN	NaN
46	96.8	6.77
47	89.9	5.11
48	108.2	7.98
49	NaN	NaN
50	96.9	5.62
51	110.3	7.01
52	NaN	NaN
53	89.5	9.14
54	105.3	6.76
55	97.0	7.84
56	NaN	NaN

	Per-Capita Health Care Spending ($)	Health Score	Adult Obesity (%)	\
0	7980	-0.75	38.9	
1	14044	-0.13	34.0	
2	8556	-0.06	33.3	
3	8562	-0.83	38.9	
4	11054	0.10	29.1	
5	8871	0.51	25.0	
6	10639	0.68	32.0	
7	11987	0.23	36.6	
8	13865	NaN	25.5	
9	9545	-0.05	29.6	
10	8680	-0.27	35.4	
11	NaN	NaN	40.2	
12	9808	0.54	27.0	
13	8078	0.26	32.7	
14	9895	-0.03	34.2	
15	11071	-0.21	38.4	
16	8660	0.24	36.6	
17	9066	0.03	37.6	

18	9964	-0.50	37.2
19	10148	-0.94	39.2
20	10913	0.47	33.2
21	9456	0.59	32.7
22	11985	0.91	27.0
23	9023	-0.08	36.1
24	10567	0.63	32.3
25	8135	-0.77	40.4
26	10036	-0.29	34.6
27	9747	-0.05	31.0
28	10192	0.20	37.6
29	7536	-0.39	34.2
30	10682	0.99	31.1
31	10468	0.51	27.7
32	8469	-0.37	34.5
33	12221	0.09	29.5
34	8744	0.13	34.5
35	11667	0.37	36.8
36	10202	-0.11	36.9
37	9052	-0.62	36.8
38	9931	0.21	33.5
39	10262	0.07	34.2
40	NaN	NaN	36.2
41	9864	0.51	31.1
42	7741	-0.18	34.6
43	12451	0.12	37.0
44	8761	-0.55	NaN
45	NaN	NaN	38.9
46	7807	-0.32	35.6
47	7233	0.64	31.0
48	11493	0.91	29.0
49	NaN	NaN	37.7
50	9123	0.40	32.3
51	9693	0.61	31.5
52	NaN	NaN	NaN
53	12055	-0.73	41.4
54	10079	0.16	37.4
55	9640	-0.04	32.5
56	NaN	NaN	34.1

	Life Expectancy (Years)
0	73.8
1	75.8
2	76.7
3	73.9
4	79.3
5	78.5

6	79.4
7	76.5
8	76.6
9	77.9
10	75.9
11	NaN
12	80.0
13	78.4
14	77.5
15	75.4
16	77.9
17	76.5
18	73.6
19	73.8
20	76.6
21	77.8
22	79.8
23	76.8
24	79.3
25	72.6
26	75.2
27	77.3
28	78.3
29	76.4
30	78.7
31	79.6
32	74.5
33	79.5
34	75.9
35	77.9
36	75.6
37	73.8
38	77.7
39	77.3
40	NaN
41	79.2
42	75.1
43	77.3
44	73.8
45	NaN
46	77.1
47	79.0
48	78.3
49	NaN
50	77.5
51	78.4
52	NaN

```
53                    72.2
54                    78.1
55                    76.8
56                     NaN
```

1.2.7 Merging and grouping split State Data (Washington and Tennessee)

[315]:
```
df_merged["State"] = df_merged["State"].str.replace("*", "", regex=False)
df_merged["State"] = df_merged["State"].replace({"Washington D.C.": "District␣
 ↪of Columbia"})

df = df_merged.groupby("State", as_index=False).first()

df.loc[8, 'Abbreviation'] = 'DC' # Adding Abbreviation to DC
```

[316]:
```
df
```

[316]:

	State	Abbreviation	Diversity Index (%)	\
0	Alabama	AL	53.1	
1	Alaska	AK	62.8	
2	Arizona	AZ	61.5	
3	Arkansas	AR	49.8	
4	California	CA	69.7	
5	Colorado	CO	52.3	
6	Connecticut	CT	55.7	
7	Delaware	DE	59.6	
8	District of Columbia	DC	67.2	
9	Florida	FL	64.1	
10	Georgia	GA	64.1	
11	Guam	NaN	NaN	
12	Hawaii	HI	76.0	
13	Idaho	ID	35.9	
14	Illinois	IL	60.3	
15	Indiana	IN	41.3	
16	Iowa	IA	30.8	
17	Kansas	KS	45.4	
18	Kentucky	KY	32.8	
19	Louisiana	LA	58.6	
20	Maine	ME	18.5	
21	Maryland	MD	67.3	
22	Massachusetts	MA	51.6	
23	Michigan	MI	45.2	
24	Minnesota	MN	40.5	
25	Mississippi	MS	55.9	
26	Missouri	MO	40.8	
27	Montana	MT	30.1	
28	Nebraska	NE	40.8	

29	Nevada	NV	68.8
30	New Hampshire	NH	23.6
31	New Jersey	NJ	65.8
32	New Mexico	NM	63.0
33	New York	NY	65.8
34	North Carolina	NC	57.9
35	North Dakota	ND	32.6
36	Ohio	OH	40.4
37	Oklahoma	OK	59.5
38	Oregon	OR	46.1
39	Pennsylvania	PA	44.0
40	Puerto Rico	NaN	NaN
41	Rhode Island	RI	49.4
42	South Carolina	SC	54.6
43	South Dakota	SD	35.6
44	Tennessee	TN	46.6
45	Texas	TX	67.0
46	Utah	UT	40.7
47	Vermont	VT	20.2
48	Virgin Islands	NaN	NaN
49	Virginia	VA	60.5
50	Washington	WA	55.9
51	West Virginia	WV	20.2
52	Wisconsin	WI	37.0
53	Wyoming	WY	32.4
54	U.S. State and Territory Average	NaN	NaN

	Healthcare Cost Index (U.S. = 100)	Health Care as % of Household Income	\
0	85.6	7.32	
1	130.7	10.08	
2	98.8	6.81	
3	85.8	7.67	
4	111.1	5.64	
5	103.9	6.05	
6	108.9	6.89	
7	101.6	6.66	
8	118.7	NaN	
9	101.4	6.05	
10	99.6	6.62	
11	NaN	NaN	
12	113.1	8.00	
13	101.6	7.82	
14	103.0	7.71	
15	91.5	6.64	
16	94.3	6.74	
17	90.6	7.19	
18	92.4	7.46	

19	91.6	8.13
20	114.7	9.30
21	106.9	6.76
22	131.8	8.00
23	88.2	6.65
24	111.5	6.09
25	85.1	9.18
26	98.0	7.43
27	106.3	8.62
28	105.6	6.87
29	105.4	5.87
30	106.5	6.42
31	115.8	5.81
32	101.9	9.07
33	108.9	7.98
34	108.4	8.78
35	107.7	8.08
36	96.5	6.68
37	98.1	7.51
38	115.3	9.32
39	95.1	6.80
40	NaN	NaN
41	104.1	6.09
42	97.6	6.13
43	104.1	8.60
44	88.7	6.77
45	96.8	6.77
46	89.9	5.11
47	108.2	7.98
48	NaN	NaN
49	96.9	5.62
50	110.3	7.01
51	89.5	9.14
52	105.3	6.76
53	97.0	7.84
54	NaN	NaN

	Per-Capita Health Care Spending ($)	Health Score	Adult Obesity (%)	\
0	7980	-0.75	38.9	
1	14044	-0.13	34.0	
2	8556	-0.06	33.3	
3	8562	-0.83	38.9	
4	11054	0.10	29.1	
5	8871	0.51	25.0	
6	10639	0.68	32.0	
7	11987	0.23	36.6	
8	13865	NaN	25.5	

9	9545	-0.05	29.6
10	8680	-0.27	35.4
11	NaN	NaN	40.2
12	9808	0.54	27.0
13	8078	0.26	32.7
14	9895	-0.03	34.2
15	11071	-0.21	38.4
16	8660	0.24	36.6
17	9066	0.03	37.6
18	9964	-0.50	37.2
19	10148	-0.94	39.2
20	10913	0.47	33.2
21	9456	0.59	32.7
22	11985	0.91	27.0
23	9023	-0.08	36.1
24	10567	0.63	32.3
25	8135	-0.77	40.4
26	10036	-0.29	34.6
27	9747	-0.05	31.0
28	10192	0.20	37.6
29	7536	-0.39	34.2
30	10682	0.99	31.1
31	10468	0.51	27.7
32	8469	-0.37	34.5
33	12221	0.09	29.5
34	8744	0.13	34.5
35	11667	0.37	36.8
36	10202	-0.11	36.9
37	9052	-0.62	36.8
38	9931	0.21	33.5
39	10262	0.07	34.2
40	NaN	NaN	36.2
41	9864	0.51	31.1
42	7741	-0.18	34.6
43	12451	0.12	37.0
44	8761	-0.55	38.9
45	7807	-0.32	35.6
46	7233	0.64	31.0
47	11493	0.91	29.0
48	NaN	NaN	37.7
49	9123	0.40	32.3
50	9693	0.61	31.5
51	12055	-0.73	41.4
52	10079	0.16	37.4
53	9640	-0.04	32.5
54	NaN	NaN	34.1

	Life Expectancy (Years)
0	73.8
1	75.8
2	76.7
3	73.9
4	79.3
5	78.5
6	79.4
7	76.5
8	76.6
9	77.9
10	75.9
11	NaN
12	80.0
13	78.4
14	77.5
15	75.4
16	77.9
17	76.5
18	73.6
19	73.8
20	76.6
21	77.8
22	79.8
23	76.8
24	79.3
25	72.6
26	75.2
27	77.3
28	78.3
29	76.4
30	78.7
31	79.6
32	74.5
33	79.5
34	75.9
35	77.9
36	75.6
37	73.8
38	77.7
39	77.3
40	NaN
41	79.2
42	75.1
43	77.3
44	73.8
45	77.1

```
46                    79.0
47                    78.3
48                     NaN
49                    77.5
50                    78.4
51                    72.2
52                    78.1
53                    76.8
54                     NaN
```

1.2.8 Final compiled CSV output

[317]:
```
df.to_csv('final_data.csv', sep=',', index=False, encoding='utf-8',␣
↪na_rep='NaN')
```

1.3 Understanding the DataFrame

[318]:
```
df.columns
```

[318]:
```
Index(['State', 'Abbreviation', 'Diversity Index (%)',
       'Healthcare Cost Index (U.S. = 100)',
       'Health Care as % of Household Income',
       'Per-Capita Health Care Spending ($)', 'Health Score',
       'Adult Obesity (%)', 'Life Expectancy (Years)'],
      dtype='str')
```

[319]:
```
df.dtypes
```

[319]:
```
State                                    str
Abbreviation                             str
Diversity Index (%)                  float64
Healthcare Cost Index (U.S. = 100)   float64
Health Care as % of Household Income     str
Per-Capita Health Care Spending ($)      str
Health Score                         float64
Adult Obesity (%)                        str
Life Expectancy (Years)              float64
dtype: object
```

1.3.1 Convert str Columns to float

[320]:
```
df['Health Care as % of Household Income'] = df['Health Care as % of Household␣
↪Income'].astype('float64')
df['Per-Capita Health Care Spending ($)'] = df['Per-Capita Health Care Spending␣
↪($)'].astype('float64')
df['Adult Obesity (%)'] = df['Adult Obesity (%)'].astype('float64')
```

[321]:
```
df.dtypes
```

[321]:
```
State                                        str
Abbreviation                                 str
Diversity Index (%)                      float64
Healthcare Cost Index (U.S. = 100)       float64
Health Care as % of Household Income     float64
Per-Capita Health Care Spending ($)      float64
Health Score                             float64
Adult Obesity (%)                        float64
Life Expectancy (Years)                  float64
dtype: object
```

[322]:
```
df.describe()
```

[322]:
```
       Diversity Index (%)  Healthcare Cost Index (U.S. = 100)  \
count                51.00                               51.00
mean                 49.40                              102.36
std                  14.60                               10.40
min                  18.50                               85.10
25%                  40.45                               95.80
50%                  51.60                              101.90
75%                  61.00                              108.30
max                  76.00                              131.80

       Health Care as % of Household Income  \
count                                 50.00
mean                                   7.29
std                                    1.13
min                                    5.11
25%                                    6.64
50%                                    6.95
75%                                    8.00
max                                   10.08

       Per-Capita Health Care Spending ($)  Health Score  Adult Obesity (%)  \
count                                51.00         50.00              55.00
mean                               9915.71          0.06              34.11
std                                1531.95          0.48               3.88
min                                7233.00         -0.94              25.00
25%                                8752.50         -0.26              31.75
50%                                9864.00          0.08              34.20
75%                               10660.50          0.45              36.95
max                               14044.00          0.99              41.40

       Life Expectancy (Years)
count                    51.00
```

```
mean                    76.88
std                      2.00
min                     72.20
25%                     75.70
50%                     77.30
75%                     78.35
max                     80.00
```

[323]:
```
df.count()
```

[323]:
```
State                                 55
Abbreviation                          51
Diversity Index (%)                   51
Healthcare Cost Index (U.S. = 100)    51
Health Care as % of Household Income  50
Per-Capita Health Care Spending ($)   51
Health Score                          50
Adult Obesity (%)                     55
Life Expectancy (Years)               51
dtype: int64
```

[324]:
```
df.isnull().sum()
```

[324]:
```
State                                 0
Abbreviation                          4
Diversity Index (%)                   4
Healthcare Cost Index (U.S. = 100)    4
Health Care as % of Household Income  5
Per-Capita Health Care Spending ($)   4
Health Score                          5
Adult Obesity (%)                     0
Life Expectancy (Years)               4
dtype: int64
```

1.4 Filling NaN with Median for data estimation

[325]:
```
numeric_cols = df.select_dtypes(include="number").columns
df[numeric_cols] = df[numeric_cols].fillna(round(df[numeric_cols].median(), 2))
```

[326]:
```
df.head()
```

[326]:

	State	Abbreviation	Diversity Index (%)	\
0	Alabama	AL	53.1	
1	Alaska	AK	62.8	
2	Arizona	AZ	61.5	
3	Arkansas	AR	49.8	
4	California	CA	69.7	

```
   Healthcare Cost Index (U.S. = 100)  Health Care as % of Household Income  \
0                                85.6                                 7.32
1                               130.7                                10.08
2                                98.8                                 6.81
3                                85.8                                 7.67
4                               111.1                                 5.64

   Per-Capita Health Care Spending ($)  Health Score  Adult Obesity (%)  \
0                               7980.0         -0.75               38.9
1                              14044.0         -0.13               34.0
2                               8556.0         -0.06               33.3
3                               8562.0         -0.83               38.9
4                              11054.0          0.10               29.1

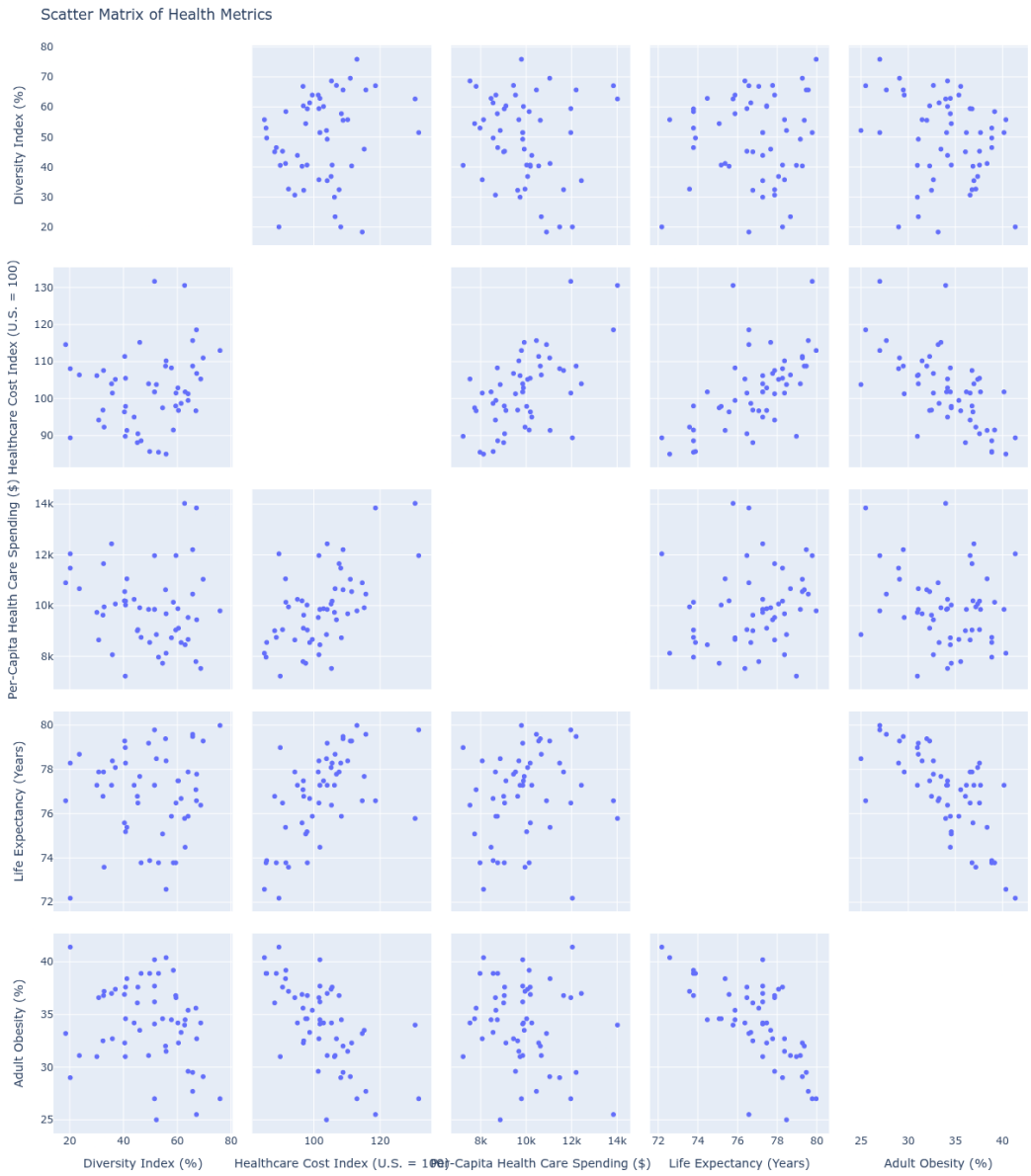
   Life Expectancy (Years)
0                     73.8
1                     75.8
2                     76.7
3                     73.9
4                     79.3
```

[327]:
```
df.to_csv('final_data(Filled NaN).csv', sep=',', index=False, encoding='utf-8')
```

2 Visualizing the Data

2.1 Exploratory Analysis

[333]:
```
fig = px.scatter_matrix(
    df,
    dimensions=[
        "Diversity Index (%)",
        "Healthcare Cost Index (U.S. = 100)",
        "Per-Capita Health Care Spending ($)",
        "Life Expectancy (Years)",
        "Adult Obesity (%)",
    ],
    hover_name="State",)
fig.update_layout(
    title=dict(text='Scatter Matrix of Health Metrics'),
    dragmode='select',
    width=1400,
    height=1400,
    hovermode='closest',
)
fig.update_traces(diagonal_visible=False)
fig.show()
```

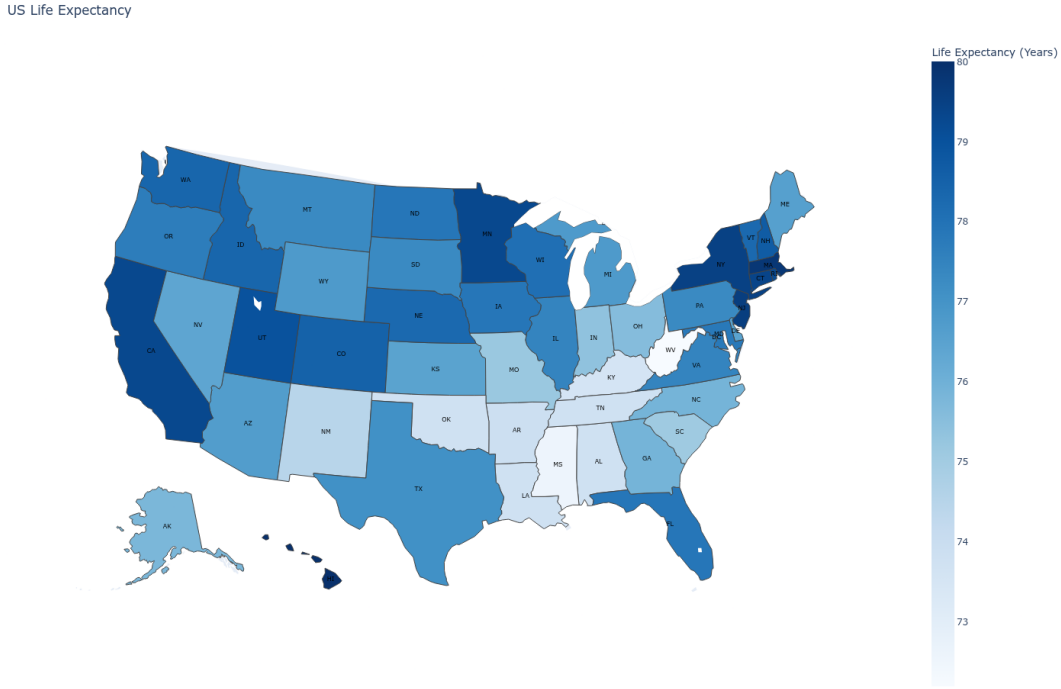


2.2 Geographical Analysis

2.2.1 USA Life Expectancy

[335]:
```
fig = px.choropleth(
    df,
    locations="State",
    locationmode="USA-states",
    color="Life Expectancy (Years)",
    scope="usa",
```

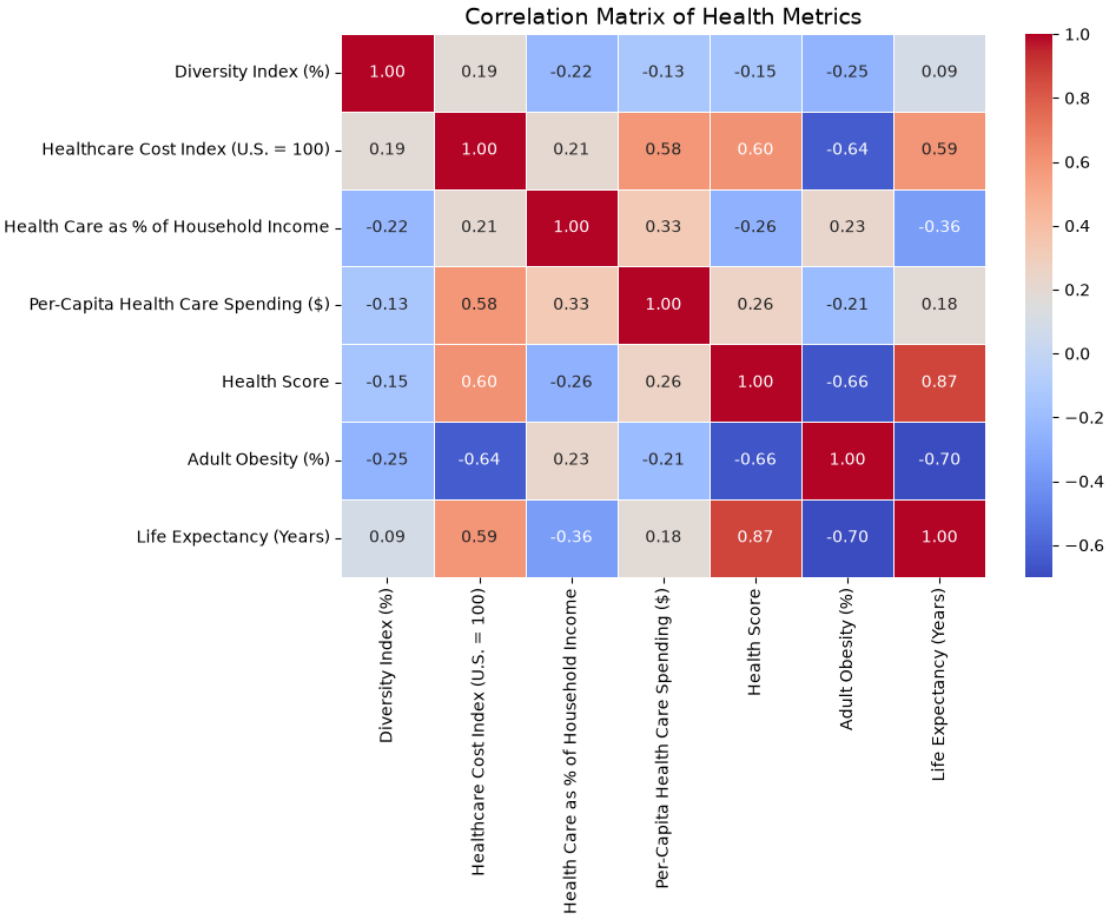
```
    color_continuous_scale="Blues",
    hover_data="Life Expectancy (Years)",
)
fig.add_scattergeo(
    locations=df['State'],
    locationmode='USA-states',
    text=df['Abbreviation'],
    mode='text',
    textfont=dict(size=8, color='black'),
    showlegend=False
)
fig.update_layout(
    title=dict(text='US Life Expectancy'),
    dragmode='select',
    width=1400,
    height=1000,
    hovermode='closest',
)
fig.show()
```



2.3 Correlation and Multivariate Relationships

[330]:
```
numeric_df = df.select_dtypes(include="number")

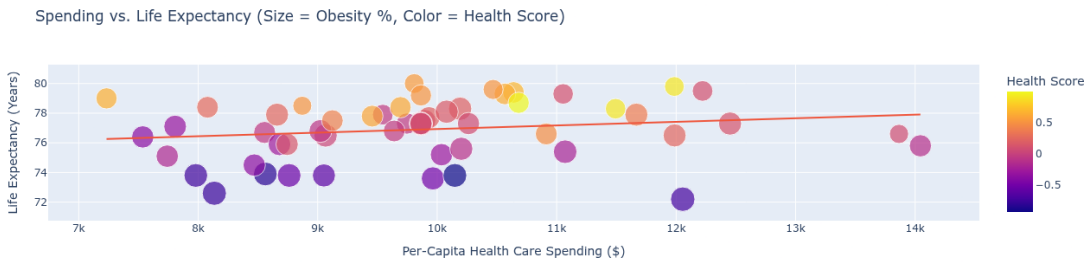
plt.figure(figsize=(10, 8))
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm", fmt=".2f",␣
 ↪linewidths=0.5)
plt.title("Correlation Matrix of Health Metrics", fontsize=14)
plt.tight_layout()
plt.show()
```



2.4 Bivariate correlation

[331]:
```
fig = px.scatter(
    df,
    x="Per-Capita Health Care Spending ($)",
    y="Life Expectancy (Years)",
    size="Adult Obesity (%)",
```

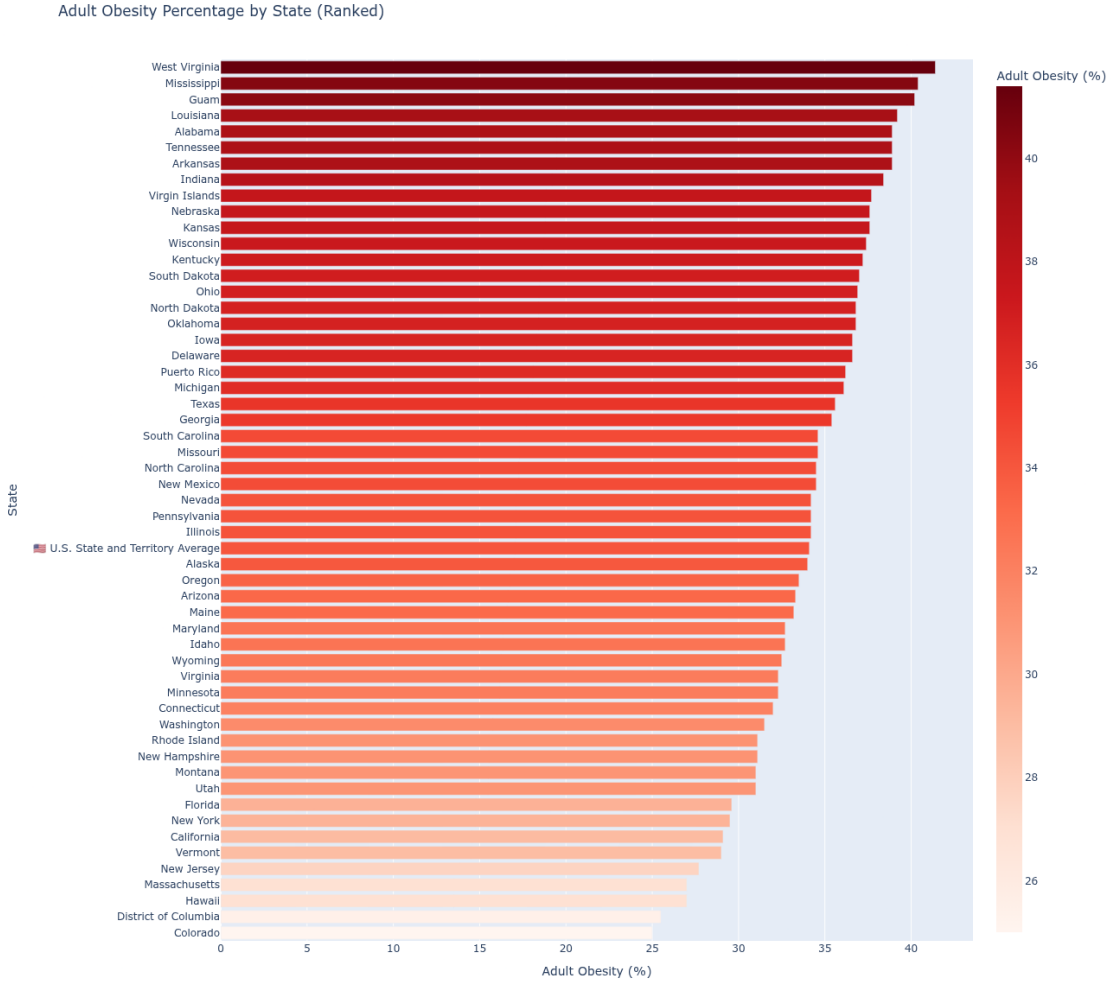
```
    color="Health Score",
    hover_name="State",
    trendline="ols",
    title="Spending vs. Life Expectancy (Size = Obesity %, Color = Health␣
 ↪Score)",
)
fig.show()
```



2.5 Categorical Comparison

[332]:
```
df_sorted = df.sort_values(by="Adult Obesity (%)", ascending=True)

fig = px.bar(
    df_sorted,
    x="Adult Obesity (%)",
    y="State",
    orientation="h",
    color="Adult Obesity (%)",
    color_continuous_scale="Reds",
    title="Adult Obesity Percentage by State (Ranked)",
)
fig.update_layout(height=1200)
fig.show()
```


Adult Obesity Percentage by State (Ranked)
West Virginia
Mississippi
Guam
Louisiana
Alabama
Tennessee
Arkansas
Indiana
Virgin Islands
Nebraska
Kansas
Wisconsin
Kentucky
South Dakota
Ohio
North Dakota
Oklahoma
Iowa
Delaware
Puerto Rico
Michigan
Texas
Georgia
South Carolina
Missouri
North Carolina
New Mexico
Nevada
Pennsylvania
Illinois
U.S. State and Territory Average
Alaska
Oregon
Arizona
Maine
Maryland
Idaho
Wyoming
Virginia
Minnesota
Connecticut
Washington
Rhode Island
New Hampshire
Montana
Utah
Florida
New York
California
Vermont
New Jersey
Massachusetts
Hawaii
District of Columbia
Colorado
State
0
5
10
15
20
25
30
35
40
Adult Obesity (%)
Adult Obesity (%)
40
38
36
34
32
30
28
26

[ ]: